Training runs: bge_m3_long ANP (cross-attention, 1 SA block, 4 heads)

Same architecture throughout; the split is the variable. Replay: random acquisition, macro-task Brier ALC.

run	split	size	context sizes, method	steps	best step	bank Brier	replay val	replay test
w64	random	w64 z32	uniform 0-31	100,000	79,700	0.1248	0.1519	0.2465
w128	random	w128 z128	uniform 0-31	100,000	97,800	0.1193	0.1485	0.2515
w64 uniform	holdout	w64 z32	uniform 0-31	100,000	1,200	0.1675	0.1735	0.2725
w64 ALC ladder	holdout	w64 z32	ladder, ALC wts	15,000	50	0.2105	not run	not run
w64 SAM	holdout	w64 z32	ladder, equal wts, SAM 0.05	30,000	5,450	0.2058	0.1667	0.2663
w64 prior-anchor	holdout	w64 z32	ladder, equal wts, anchor	10,000	100	0.2066	not run	not run
w64 SAM, ALC ladder	holdout	w64 z32	ladder, ALC wts, SAM 0.05	30,000	4,350	0.2063	not run	not run
w64 dense-prefix	holdout	w64 z32	every prefix	10,000	7,800	0.1694	0.1735	0.2762

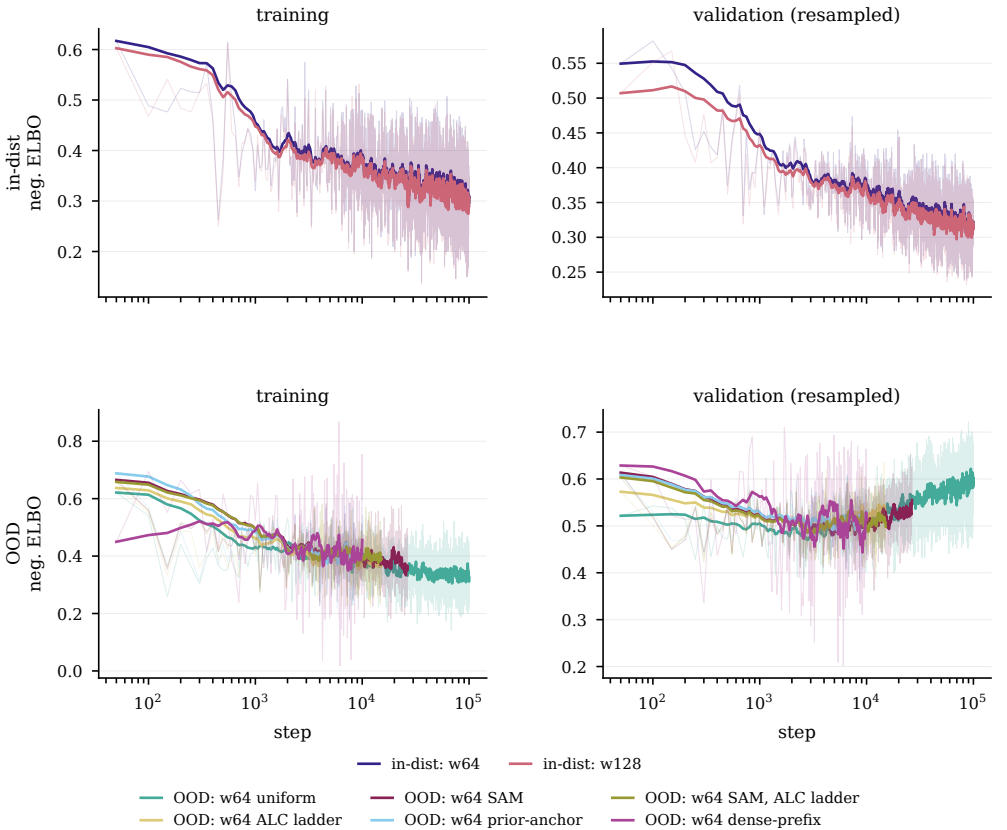
bank Brier: best value on the run's own fixed validation bank, the metric that picked its checkpoint. That bank's context sizes follow the run's sampling scheme, and in-distribution banks hold benchmarks the run trained on, so bank Brier compares only runs with the same bank.

replay val / test: random acquisition on the 82-task validation and 45-task test panels, the same protocol for every row. The random-split runs trained on ai2d_test and bird_sql (validation) and editbench (test).

dense-prefix kept training after its replay and CodaBench checkpoint was frozen at step 4,550 (bank Brier 0.1700); its final best is the step shown. SAM ran to 30,000 steps but its best checkpoint, step 5,450, did not change.

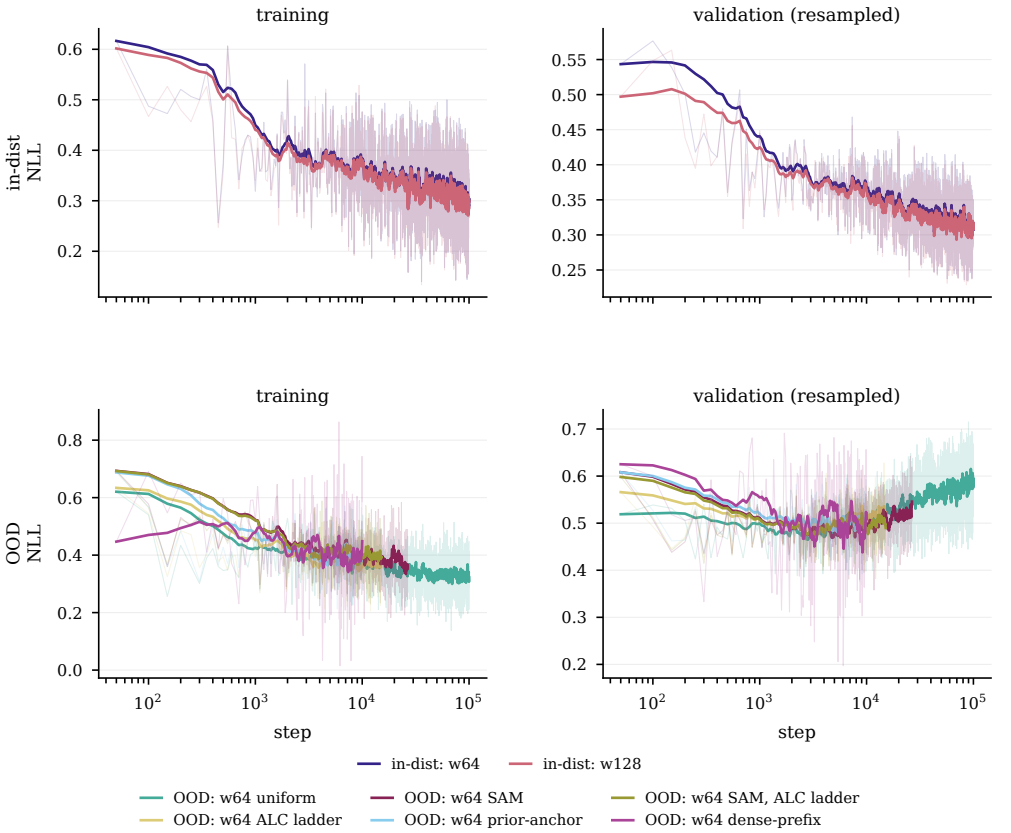
Objective: negative ELBO per target

x: optimiser step (log scale). Faint: logged values; solid: exponential moving average (span 20 points).



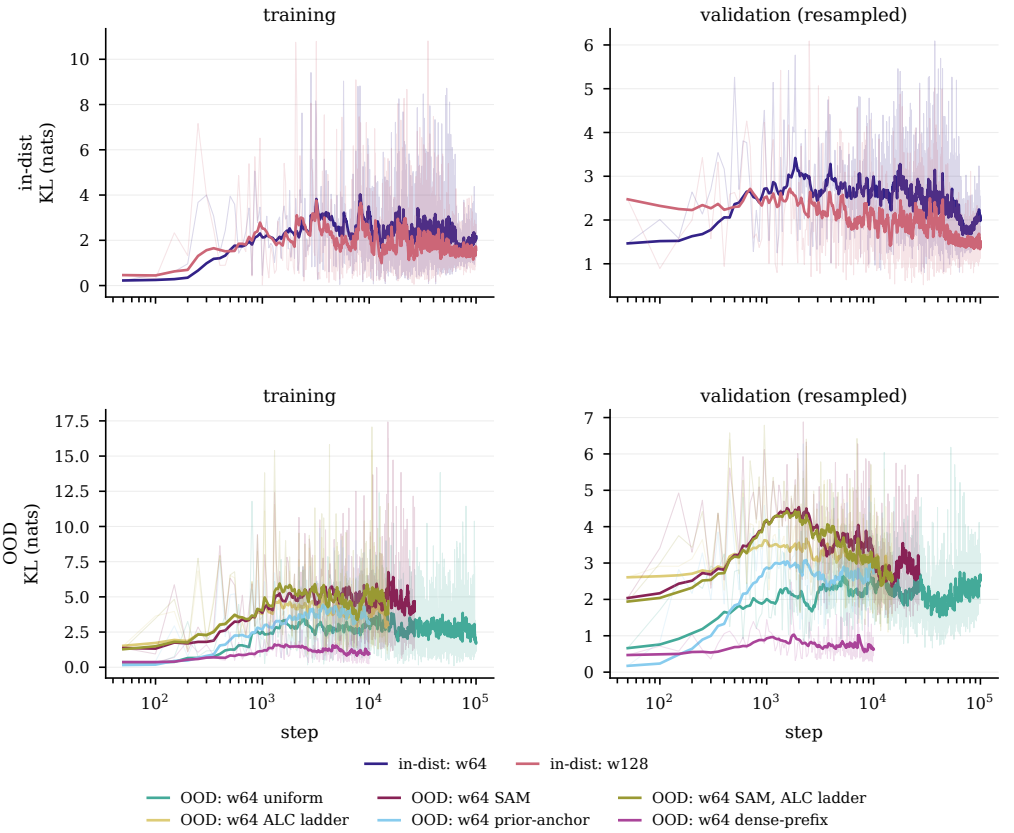
Posterior target NLL

x: optimiser step (log scale). Faint: logged values; solid: exponential moving average (span 20 points).



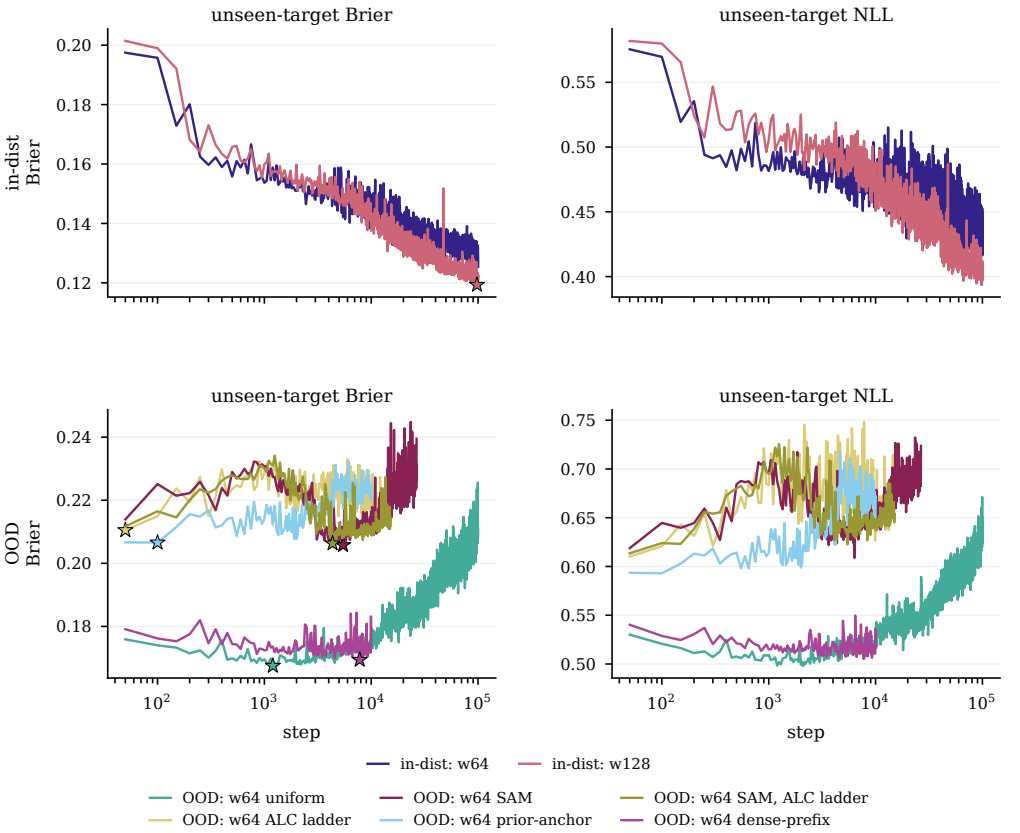
Latent KL (total)

x: optimiser step (log scale). Faint: logged values; solid: exponential moving average (span 20 points).



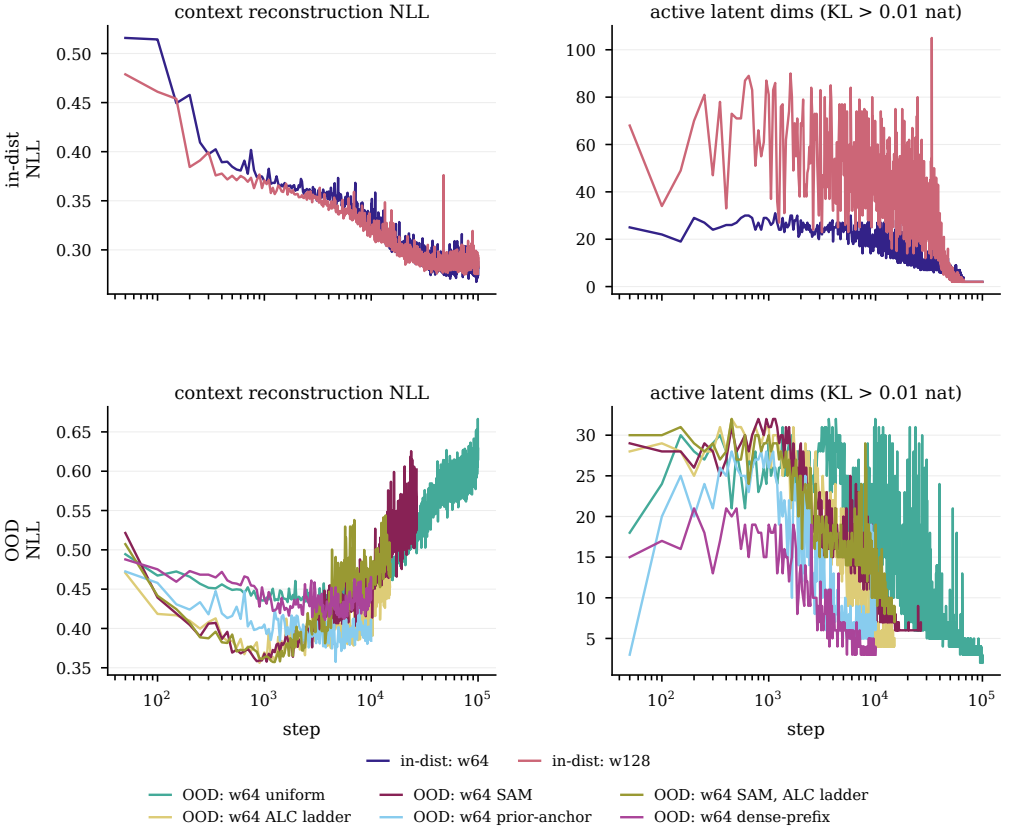
Fixed validation bank: checkpoint-selection metric

x: optimiser step (log scale). Star: the step each run's checkpoint was selected at.



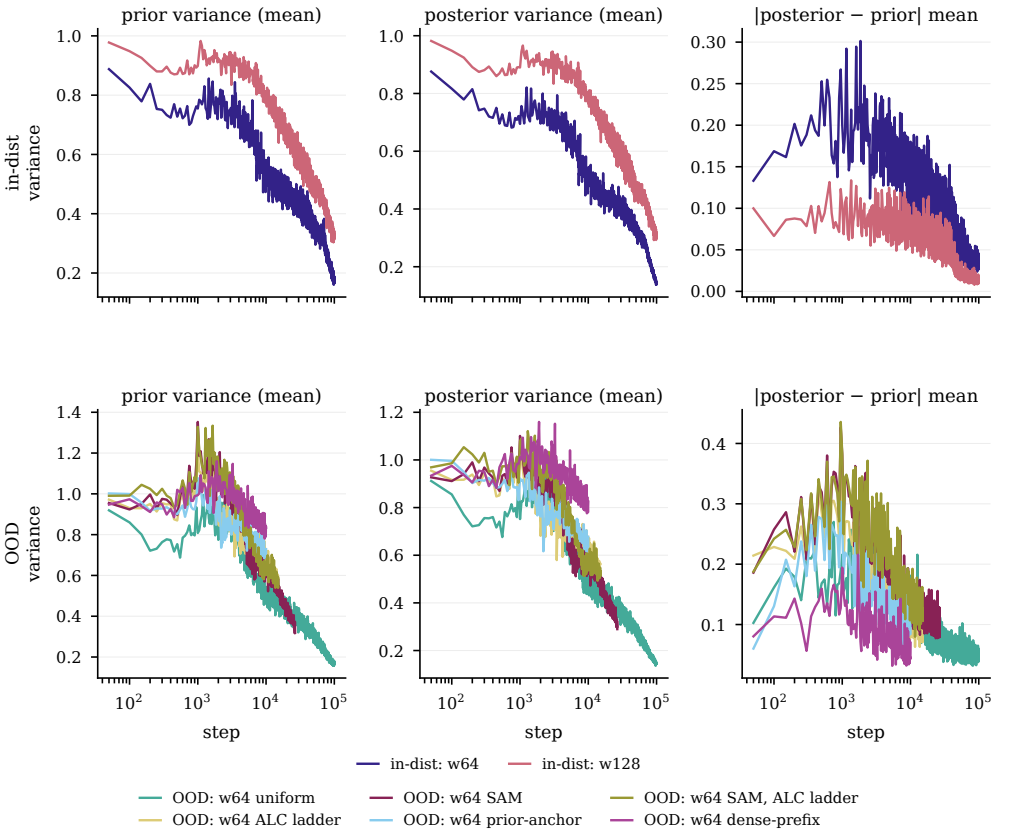
Fixed validation bank: context fit and latent use

x: optimiser step (log scale). Values as logged.



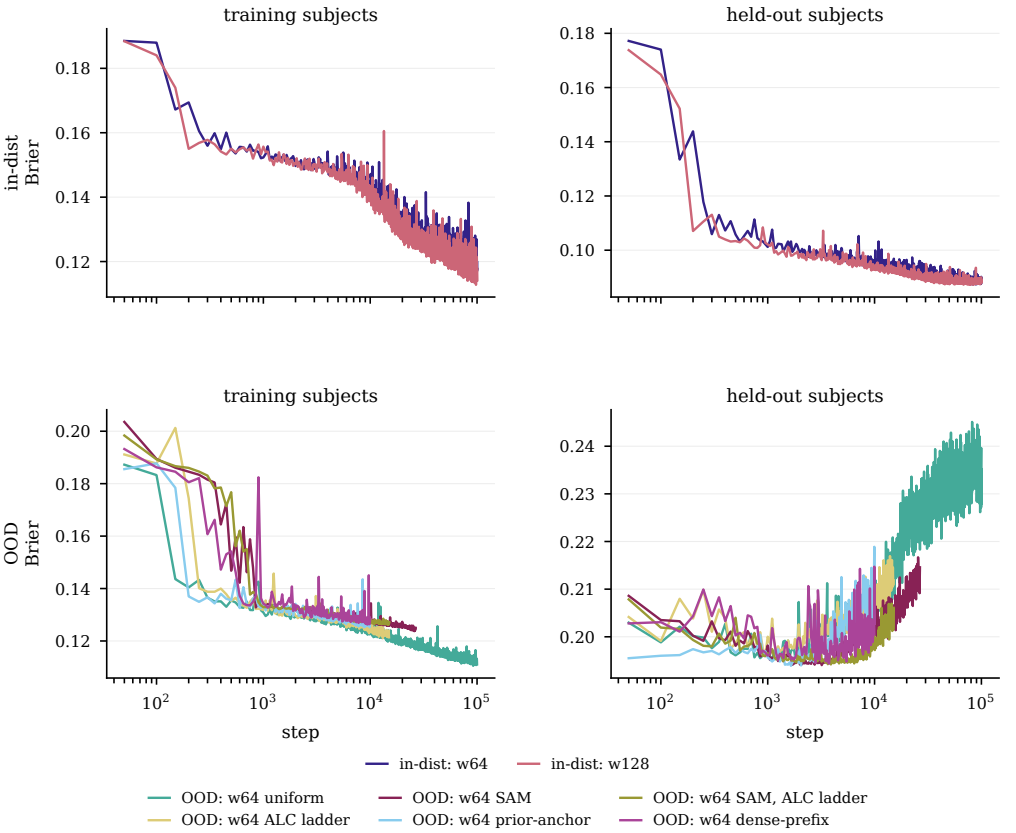
Latent distribution

x: optimiser step (log scale). Values as logged.



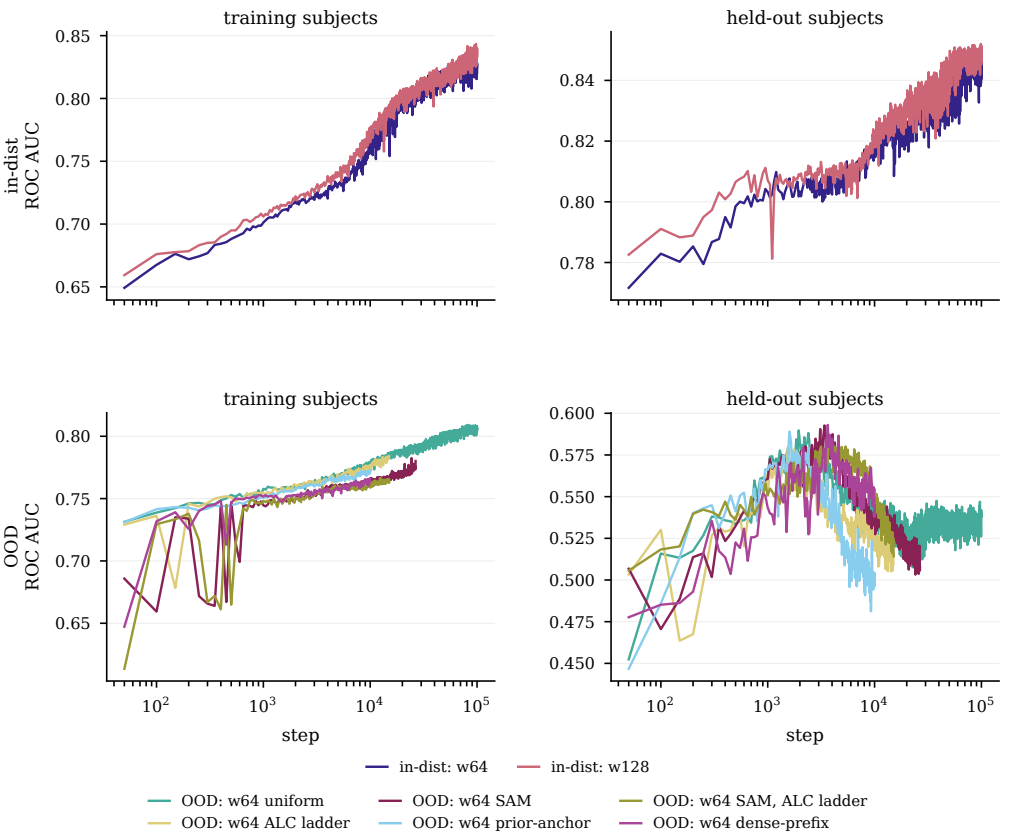
Generalisation: same-subject split-half Brier

x: optimiser step (log scale). Values as logged.



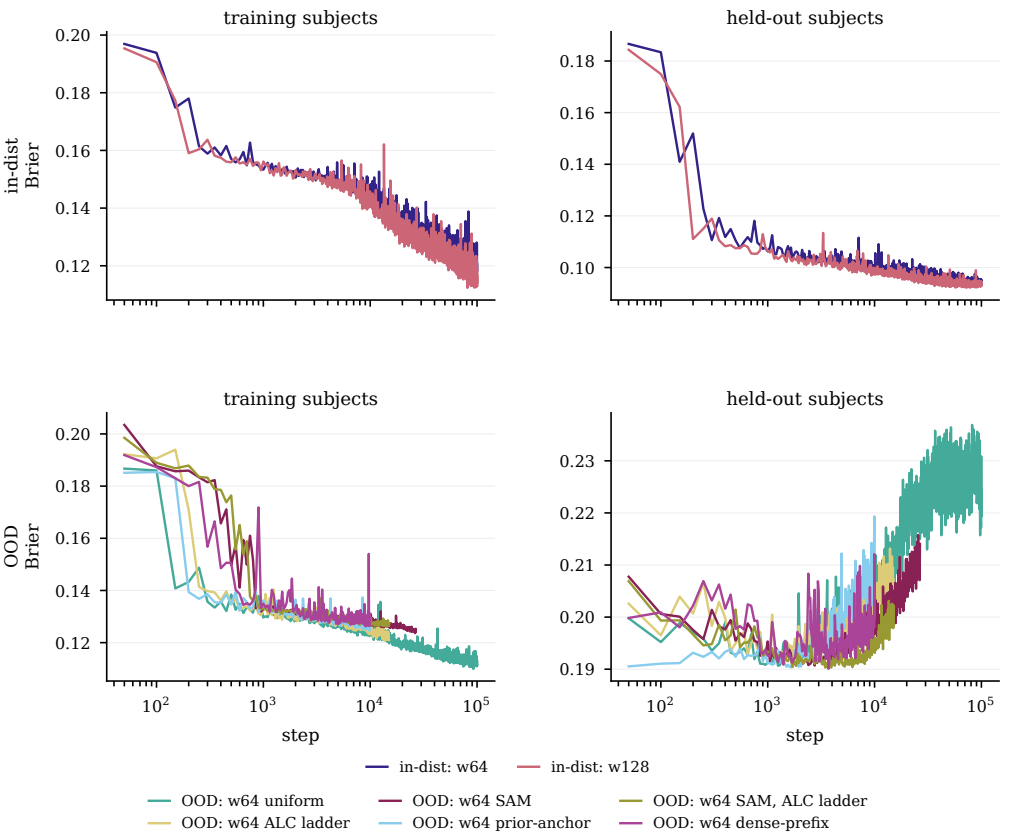
Generalisation: same-subject split-half AUC

x: optimiser step (log scale). Values as logged.



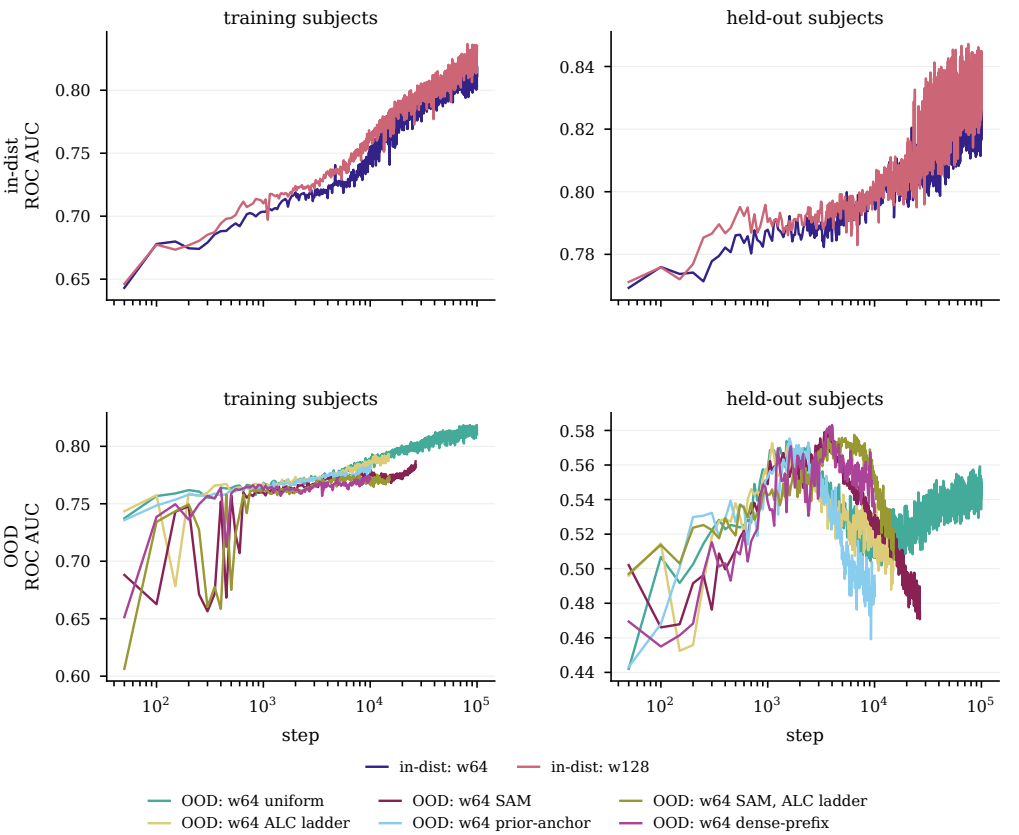
Generalisation: same-subject response-holdout Brier

x: optimiser step (log scale). Values as logged.



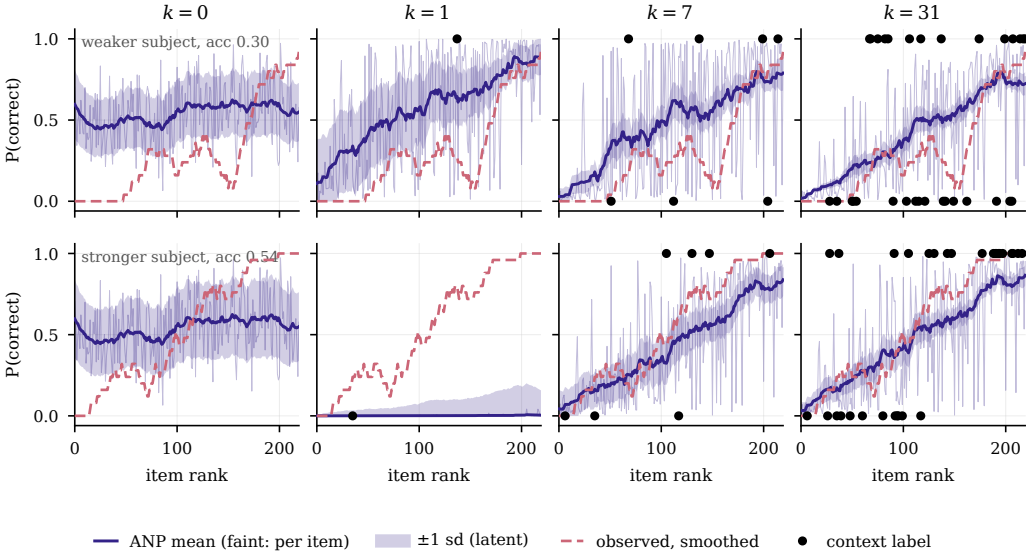
Generalisation: same-subject response-holdout AUC

x: optimiser step (log scale). Values as logged.



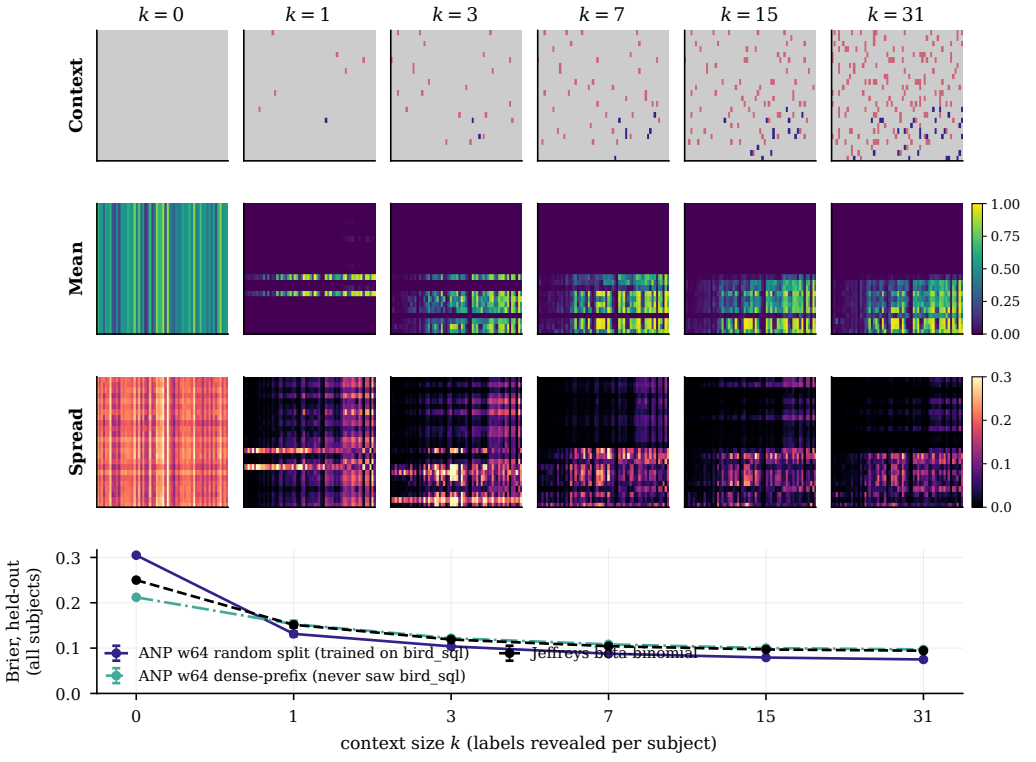
ANP learning one subject from context (in distribution)

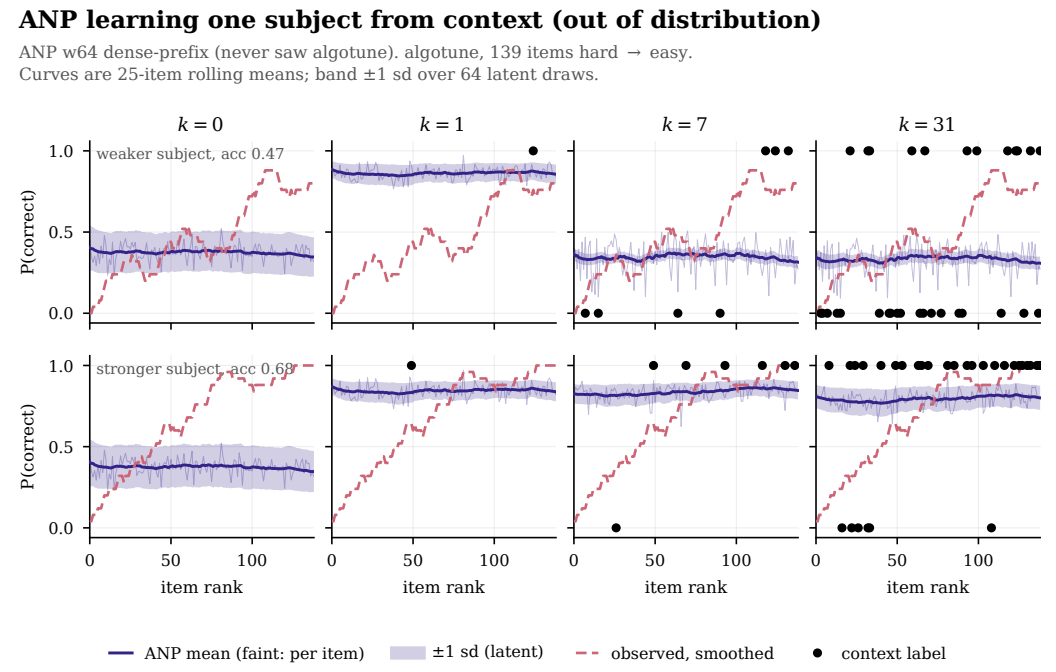
ANP w64 random split (trained on bird_sql). bird_sql, 220 items hard → easy.
Curves are 25-item rolling means; band ±1 sd over 64 latent draws.



Response-matrix completion by the ANP (in distribution)

bird_sql: 24 subjects (weak → strong) x 60 items (hard → easy). Random context order; grey = unrevealed.




ANP learning one subject from context (out of distribution)
ANP w64 dense-prefix (never saw algotune). algotune, 139 items hard → easy.
Curves are 25-item rolling means; band ±1 sd over 64 latent draws.
k = 0
k = 1
k = 7
k = 31
weaker subject, acc 0.47
stronger subject, acc 0.68
P(correct)
1.0
0.5
0.0
0
50
100
item rank
ANP mean (faint: per item)
±1 sd (latent)
observed, smoothed
context label

Response-matrix completion by the ANP (out of distribution)

algotune: 18 subjects (weak → strong) x 60 items (hard → easy). Random context order; grey = unrevealed.

